QUANTITATIVE ANALYSIS OF EMPLOYEE ATTRITION USING TABLEAU

Rati Srivastava

DATA VISUALISATION (EXPLORATORY COURSE)

1. Introduction

This study analyzes employee attrition using a dataset of 1,470 employees through an interactive Tableau dashboard to identify key factors influencing workforce turnover. The overall attrition rate is approximately 16%, indicating that nearly one in six employees leaves the organization. The analysis highlights overtime as the most significant driver, with employees working overtime showing substantially higher attrition rates. Attrition is also concentrated within the first five years of employment and among lower income groups and high-pressure roles such as sales and technical positions. Through risk segmentation, employees were categorized into low, medium, and high-risk groups to enable targeted insights. The findings demonstrate that attrition is driven by a combination of workload, compensation, and role-specific pressures, and that data-driven strategies can effectively improve employee retention.

Employee attrition is not merely a human resource concern but a measurable indicator of organizational health, directly influencing productivity, operational continuity, and financial performance. Each employee exit carries both direct costs, such as recruitment and training, and indirect costs, including loss of domain knowledge, reduced team efficiency, and disruption of workflow. When attrition reaches moderate levels such as the ~16% observed in this dataset, it signals underlying inefficiencies rather than isolated or random departures. In such cases, understanding attrition becomes less about counting exits and more about diagnosing the structural conditions that lead to them.

From an analytical perspective, attrition represents a multi-variable problem where factors such as workload, compensation, job role, tenure, and work-life balance interact in non-linear ways. Traditional reporting methods often fail to capture these interactions, presenting aggregated metrics that mask critical patterns. As a result, organizations risk implementing broad, ineffective policies instead of targeted interventions. This creates a clear need for a data-driven approach that can reveal hidden relationships and quantify the relative impact of different factors.

The purpose of this study is to address this gap by analyzing workforce data through an interactive Tableau dashboard, enabling a deeper exploration of how various employee attributes influence attrition. Rather than treating attrition as a single outcome, the analysis seeks to uncover patterns across departments, roles, and behavioral indicators, and to examine how these patterns shift under different conditions. The dashboard structure supports this by combining summary indicators with comparative and distribution-based visualizations, allowing both high-level overview and detailed investigation.

To support this analysis, key visual elements such as **KPI summary panels, departmental comparisons, role-based distributions, and behavioral impact charts** are integrated throughout the report.

Figure 1: Full Dashboard Overview

Type	Field Name	Physical Table	Remote Field Name
Abc	Department	HR Data.xlsx - HR data.csv	Department
Abc	Education Field	HR Data.xlsx - HR data.csv	Education Field
Abc	Emp No	HR Data.xlsx - HR data.csv	emp no
#	Employee Number	HR Data.xlsx - HR data.csv	Employee Number
Abc	Gender	HR Data.xlsx - HR data.csv	Gender
Abc	Job Role	HR Data.xlsx - HR data.csv	Job Role
Abc	Marital Status	HR Data.xlsx - HR data.csv	Marital Status
Abc	Over Time	HR Data.xlsx - HR data.csv	Over Time

Figure 2: Key Workforce Metrics

These visual components serve as analytical anchors, ensuring that observations are grounded in quantifiable evidence rather than descriptive assumptions. Through this approach, the study aims to move beyond surface-level reporting and establish a structured understanding of employee attrition as a predictable and analyzable system.

2. Dataset and Problem Definition

The analysis is based on a structured HR dataset comprising **1,470 employee records**, where each record represents an individual employee along with a combination of demographic, professional, and behavioral attributes. The dataset includes both **categorical variables**, such as department, job role, gender, overtime status, and attrition (Yes/No), as well as **numerical variables**, including monthly income, years at the company, age, and other performance-related indicators. This combination enables both comparative and distribution-based analysis, allowing patterns to be observed across different dimensions of the workforce.

Within this dataset, **attrition is defined as a binary outcome variable**, indicating whether an employee has left the organization ("Yes") or is still retained ("No"). Out of the total 1,470 employees, **237**

instances correspond to attrition, resulting in an overall attrition rate of approximately **16.12%**. This variable serves as the central point of analysis, with all other features evaluated in terms of their relationship to employee exit behavior.

To provide a structured overview of the dataset composition and key metrics, the following summary elements are included:

Table 1: Structured overview of the dataset composition

Key Workforce Metrics	1470
Attrition Count	237
Attrition Rate	16.32%

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA
1	Attrition	loyee Nun	emp no	Gender	Job Role	Over Times	Satisfactis	At Comp	Education	ment Satis	Hourly Rate	Job Level	Satisfact	nthly Inco	onthly Rat	mpanies	ent Salary	ormance	Rnship Satis	andard Hoi	Option Li	Working	Yk Life Bala	s At Comp	In Currenhce	Last Pr	With Curr N
2	Yes	1	STAFF-1	Female	les Executi	Yes	4	6	ociates Dej	2	94	2	4	5993	19479	8	11	3	1	80	0	8	1	6	4	0	5
3	No	2	STAFF-2	Male	earch Scie	No	2	10	High Schoo	3	61	2	2	5130	24907	1	23	4	4	80	1	10	3	10	7	1	7
4	Yes	4	STAFF-4	Male	atory Tech	Yes	3	0	ociates Dej	4	92	1	3	2090	2396	6	15	3	2	80	0	7	3	0	0	0	0
5	No	5	STAFF-5	Female	earch Scie	Yes	3	8	ster's Degi	4	56	1	3	2909	23159	1	11	3	3	80	0	8	3	8	7	3	0
6	No	7	STAFF-7	Male	atory Tech	No	2	2	High Schoo	1	40	1	2	3468	16632	9	12	3	4	80	1	6	3	2	2	2	2
7	No	8	STAFF-8	Male	atory Tech	No	4	7	ociates Dej	4	79	1	4	3068	11864	0	13	3	3	80	0	8	2	7	7	3	6
8	No	10	STAFF-10	Female	atory Tech	Yes	1	1	helor's Deg	3	81	1	1	2670	9964	4	20	4	1	80	3	12	2	1	0	0	0
9	No	11	STAFF-11	Male	atory Tech	No	3	1	High Schoo	4	67	1	3	2693	13335	1	22	4	2	80	1	1	3	1	0	0	0
10	No	12	STAFF-12	Male	acturing D	No	3	9	helor's Deg	4	44	3	3	9526	8787	0	21	4	2	80	0	10	3	9	7	1	8
11	No	13	STAFF-13	Male	ire Repres	No	3	7	helor's Deg	3	94	2	3	5237	16577	6	13	3	2	80	2	17	2	7	7	7	7
12	No	14	STAFF-14	Male	atory Tech	No	2	5	helor's Deg	1	84	1	2	2426	16479	0	13	3	3	80	1	6	3	5	4	0	3
13	No	15	STAFF-15	Female	atory Tech	Yes	3	9	ociates Dej	4	49	2	3	4193	12682	0	12	3	4	80	0	10	3	9	5	0	8
14	No	16	STAFF-16	Male	earch Scie	No	3	5	High Schoo	1	31	1	3	2911	15170	1	17	3	4	80	1	5	2	5	2	4	3
15	No	18	STAFF-18	Male	atory Tech	No	4	2	ociates Dej	2	93	1	4	2661	8758	0	11	3	3	80	1	3	3	2	2	1	2
16	Yes	19	STAFF-19	Male	atory Tech	Yes	3	4	helor's Deg	3	50	1	3	2028	12947	5	14	3	2	80	0	6	3	4	2	0	3
17	No	20	STAFF-20	Female	acturing D	No	1	10	ster's Degi	2	51	3	1	9980	10195	1	11	3	3	80	1	10	3	10	9	8	8
18	No	21	STAFF-21	Male	earch Scie	Yes	2	6	ociates Dej	1	80	1	2	3298	15053	0	12	3	4	80	2	7	2	6	2	0	5
19	No	22	STAFF-22	Male	atory Tech	Yes	4	1	ociates Dej	4	96	1	4	2935	7324	1	13	3	2	80	2	1	2	1	0	0	0
20	No	23	STAFF-23	Female	Manager	No	4	25	ster's Degi	1	78	4	4	15427	22021	2	16	3	3	80	0	31	3	25	8	3	7
21	No	24	STAFF-24	Male	earch Scie	Yes	4	3	helor's Deg	4	45	1	4	3944	4306	5	11	3	3	80	0	6	3	3	2	1	2
22	No	26	STAFF-26	Female	acturing D	No	3	4	ociates Dej	1	96	2	3	4011	8232	0	18	3	4	80	1	5	2	4	2	1	3
23	Yes	27	STAFF-27	Male	Represent	No	1	5	ster's Degi	3	82	1	1	3407	6986	7	23	4	2	80	0	10	3	5	3	0	3
24	No	28	STAFF-28	Female	earch Dire	No	2	12	ster's Degi	1	53	3	2	11994	21293	0	11	3	3	80	0	13	3	12	6	2	11
25	No	30	STAFF-30	Male	earch Scie	No	4	0	ociates Dej	3	96	1	4	1232	19281	1	14	3	4	80	0	0	3	0	0	0	0
26	Yes	31	STAFF-31	Male	earch Scie	No	1	4	High Schoo	2	83	1	1	2960	17102	2	11	3	3	80	0	8	3	4	2	1	3
27	No	32	STAFF-32	Female	Manager	No	3	14	helor's Deg	3	58	5	3	19094	10735	4	11	3	4	80	1	26	2	14	13	4	8
28	Yes	33	STAFF-33	Female	earch Scie	Yes	1	10	High Schoo	2	72	1	1	3919	4681	1	22	4	2	80	0	10	3	10	2	6	7
29	No	35	STAFF-35	Male	les Executi	No	2	9	ster's Degi	3	48	2	2	6825	21173	0	11	3	4	80	1	10	3	9	7	4	2
30	No	36	STAFF-36	Female	ire Repres	No	4	22	ster's Degi	1	42	3	4	10248	2094	3	14	3	4	80	1	24	3	22	6	5	17
31	No	38	STAFF-38	Female	Manager	No	1	2	ster's Degi	2	83	5	1	18947	22822	3	12	3	4	80	0	22	2	2	2	2	1
32	No	39	STAFF-39	Male	atory Tech	No	4	1	helor's Deg	3	78	1	4	2496	6670	4	11	3	4	80	0	7	3	1	1	0	0
33	No	40	STAFF-40	Male	ire Repres	Yes	4	4	ster's Degi	4	41	2	4	6465	19121	2	13	3	4	80	0	9	4	4	2	1	3
34	No	41	STAFF-41	Male	atory Tech	No	3	10	ociates Dej	4	83	1	3	2206	16117	1	13	3	1	80	0	10	3	10	0	1	8
35	Yes	42	STAFF-42	Male	Represent	No	4	1	helor's Deg	4	56	2	4	2086	3335	3	14	3	3	80	1	19	4	1	0	0	0
36	Yes	45	STAFF-45	Male	earch Scie	Yes	4	2	helor's Deg	2	61	1	4	2293	3020	2	16	3	1	80	1	6	2	2	0	2	0

Figure 3: Initial Data Distribution or Data Overview Chart

The core analytical problem addressed in this study is not simply to report attrition levels, but to **identify and quantify the factors that significantly influence attrition and to understand how these factors interact across different segments of the organization**. This involves examining how variables such as workload (overtime), compensation (monthly income), job structure (role and department), and experience (tenure) contribute individually and collectively to attrition outcomes.

A key challenge in this context is that these variables do not operate independently. For instance, the effect of overtime may be more pronounced in certain departments, or income-related attrition may vary across job roles. Therefore, the analysis must move beyond isolated comparisons and instead explore **multi-dimensional relationships**, where patterns emerge through the interaction of multiple factors.

To support this, visual representations such as **categorical comparisons, distribution plots, and segmented dashboards** are incorporated throughout the report.

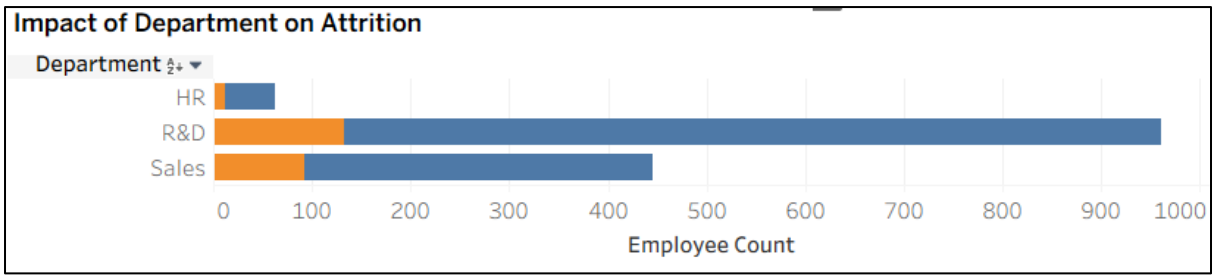

Figure 4: Sample Comparative Chart (e.g., Department vs Attrition)

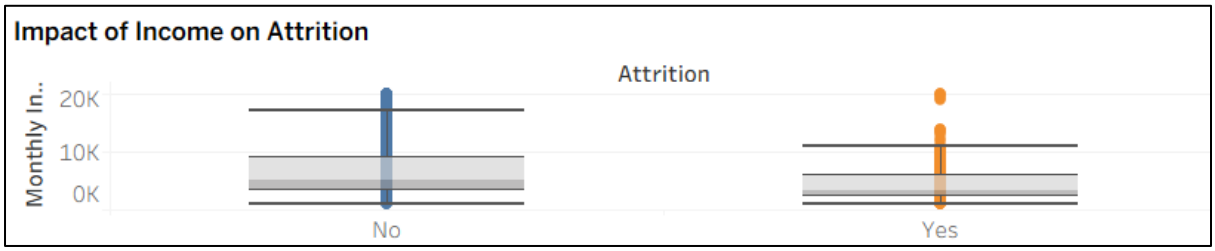

Figure 5: Sample Distribution Plot (e.g., Income vs Attrition)

By structuring the problem in this way, the dataset is treated not just as a collection of records, but as a system where employee behavior can be analyzed, interpreted, and ultimately predicted through data-driven methods.

3. Methodology and Dashboard Design

The transformation of raw HR data into analytical insights was carried out within Tableau through a structured process that ensured both data integrity and analytical clarity. The dataset was first imported and examined for consistency in data types, where categorical variables such as department, job role, attrition status, and overtime were properly assigned as dimensions, while numerical variables like monthly income and years at company were treated as measures. This step ensured that subsequent aggregations and visualizations reflected accurate relationships rather than misclassified data.

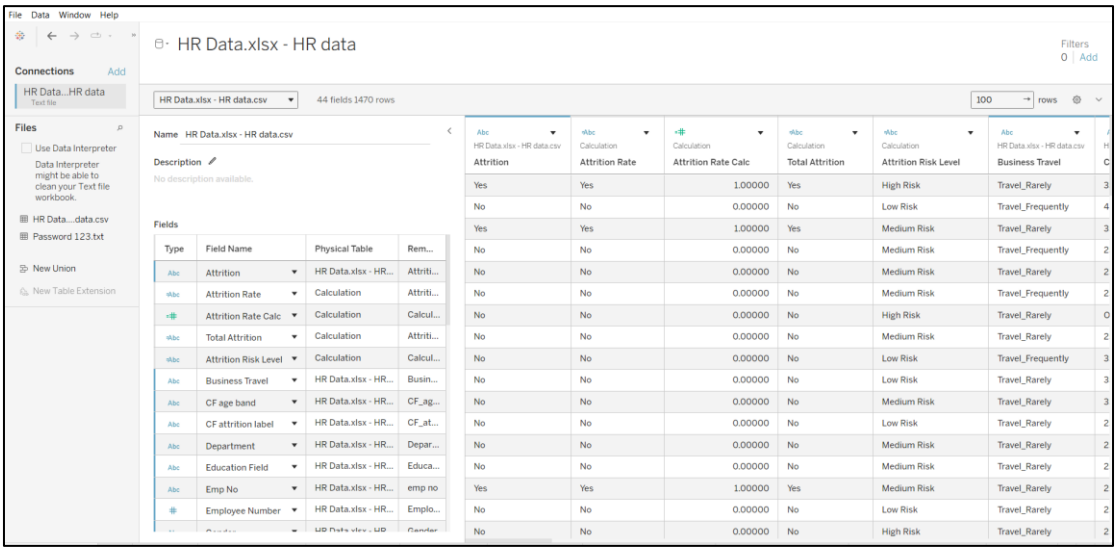

Figure 6: Data Source and Field Classification View in Tableau

To move beyond raw counts and enable meaningful interpretation, calculated fields were created. The most fundamental among these was the **attrition rate**, derived as the proportion of employees with attrition status marked "Yes" relative to total employees. In addition, a **risk segmentation field** was developed to classify employees into low, medium, and high-risk categories based on combinations of overtime, work-life balance, income level, and tenure. These calculated fields allowed the analysis to shift from descriptive visualization to structured evaluation of employee risk profiles.

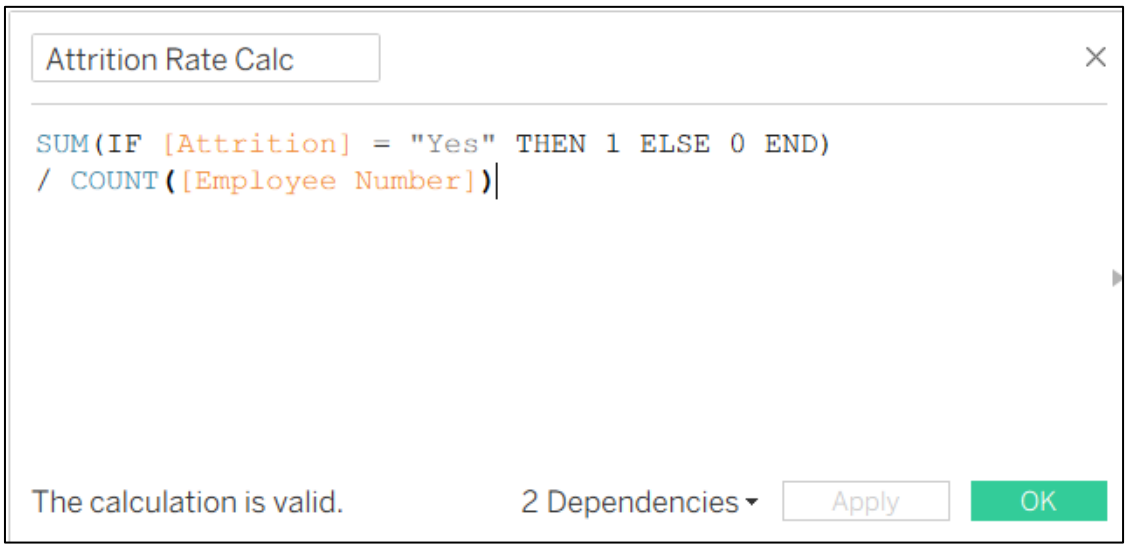

Figure 7: Calculated Field – Attrition Rate

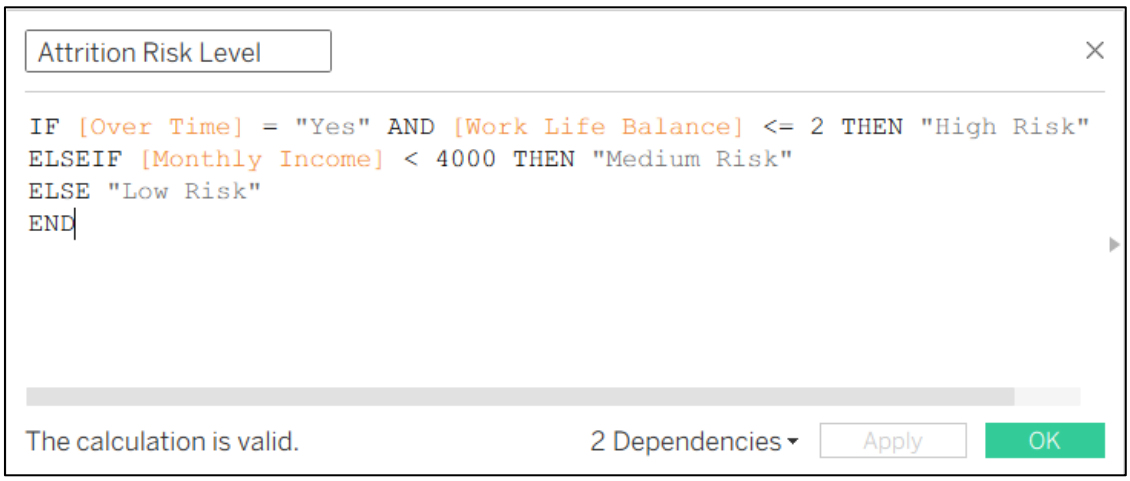

Figure 8: Calculated Field – Risk Segmentation Logic

To better capture patterns within continuous variables, binning techniques were applied. In particular, **years at company** was grouped into tenure bands (such as 0–5 years, 5–10 years, and above 10 years), enabling clearer observation of attrition concentration across different stages of employment. This approach reduced noise from raw numerical dispersion and highlighted critical retention windows. The dashboard design was guided by the need to balance overview and depth. Summary KPIs were placed at the top to establish a baseline, followed by comparative charts such as department-wise and role-wise attrition to identify structural differences. Distribution plots, including income and tenure, were incorporated to reveal underlying spread and clustering, while behavioral indicators like overtime and work-life balance were visualized to capture causal signals. Each chart type was selected based on the nature of the variable—categorical comparisons were represented through bar charts for clarity, while distributions were shown using box plots or grouped bars to preserve variability.

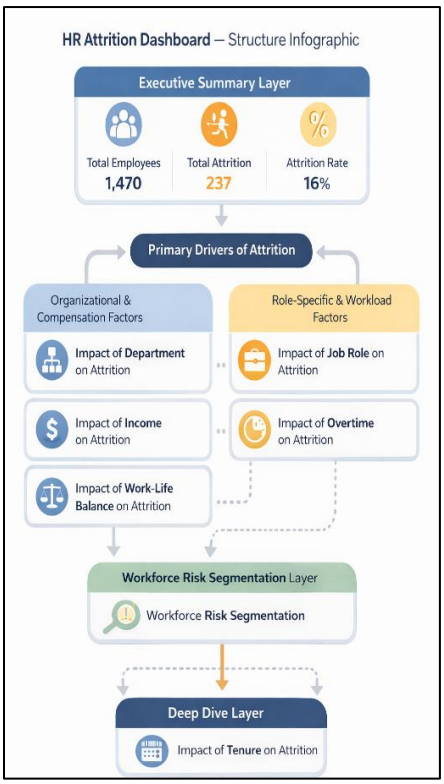

Figure 9: Dashboard Layout Structure

This design is further enhanced by the integration of interactive filters, allowing the dashboard to function as a multi-dimensional analytical system rather than a static display. Filters based on **gender, department, and job role** enable the same dataset to be viewed under different conditions, revealing how attrition patterns shift across workforce segments. This capability allows for targeted exploration, where insights can be refined from general trends to segment-specific behavior without restructuring the underlying analysis.

Figure 10: Dashboard with Active Filters Applied

Through this methodology, the raw dataset is systematically transformed into a layered analytical framework, where calculated metrics, structured grouping, and interactive design collectively enable deeper understanding of employee attrition.

4. Dashboard Overview

The Tableau dashboard is designed as an integrated analytical system that consolidates multiple dimensions of workforce data into a single, interactive interface. Rather than presenting isolated charts, the dashboard is structured to guide the user from a high-level understanding of attrition toward progressively deeper insights across organizational, financial, and behavioral variables.

Figure 11: Full Dashboard Overview

At the top of the dashboard, a set of key performance indicators establishes the baseline context by displaying total employees, total attrition, and overall attrition rate. This section anchors the analysis, allowing all subsequent observations to be interpreted relative to the overall workforce condition.

Below this, comparative visualizations such as department-wise and job role-wise attrition are positioned to highlight structural differences within the organization. These charts enable immediate identification of segments where attrition is disproportionately high, making them essential for isolating high-risk areas.

The dashboard further incorporates distribution-based visualizations, including income and tenure, to capture how attrition varies across continuous variables. These charts provide depth by revealing concentration patterns, such as whether attrition is clustered within specific income ranges or early stages of employment.

Behavioral indicators, particularly overtime and work-life balance, are included to represent operational and experiential factors influencing employee decisions. These visual elements are critical in linking organizational practices with attrition outcomes.

Finally, a risk segmentation component categorizes employees into distinct risk groups, synthesizing multiple variables into a single interpretable framework. This section supports targeted analysis by identifying which portions of the workforce are most vulnerable to attrition.

Overall, the dashboard is structured to move from summary to detail, enabling both immediate interpretation and deeper exploration, while maintaining a coherent analytical flow across all components.

5. Quantitative Analysis of Attrition Drivers

5.1. Department and Role-Based Structural Influence

The analysis of attrition drivers begins with organizational structure, where department-level comparisons provide the first indication of how turnover is distributed across the workforce. The department chart represents each unit through stacked bars, where the total length corresponds to employee count and the highlighted segment reflects attrition.

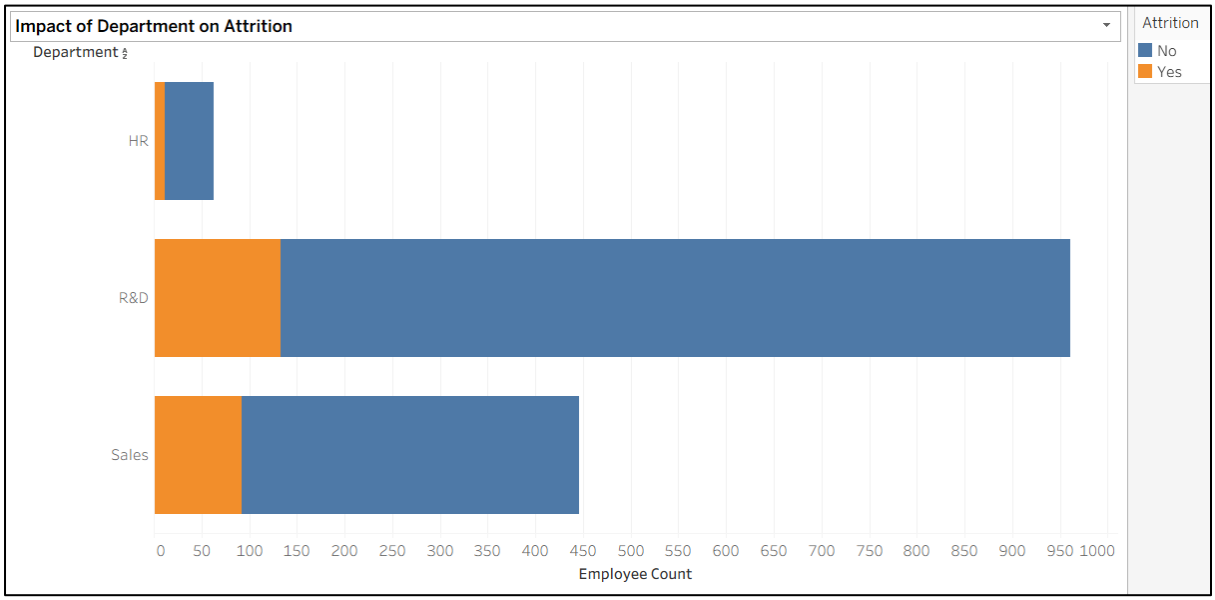

Figure 12: Department vs Attrition Chart

Within this visualization, Research & Development accounts for the largest share of employees (approximately 961), with around 133 attrition cases, resulting in an attrition rate close to **13.8%**, which is below the overall average of 16.12%. In contrast, the Sales department, with roughly 446 employees and about 92 attrition cases, exhibits a significantly higher attrition rate of approximately **20.6%**. The visual distinction is evident as the attrition segment occupies a larger proportion of the total bar in Sales compared to R&D, indicating that attrition is not merely a function of workforce size but is influenced by the operational intensity and performance-driven nature of certain departments.

Extending this analysis to the job role level provides a more granular understanding of how attrition is distributed within departments.

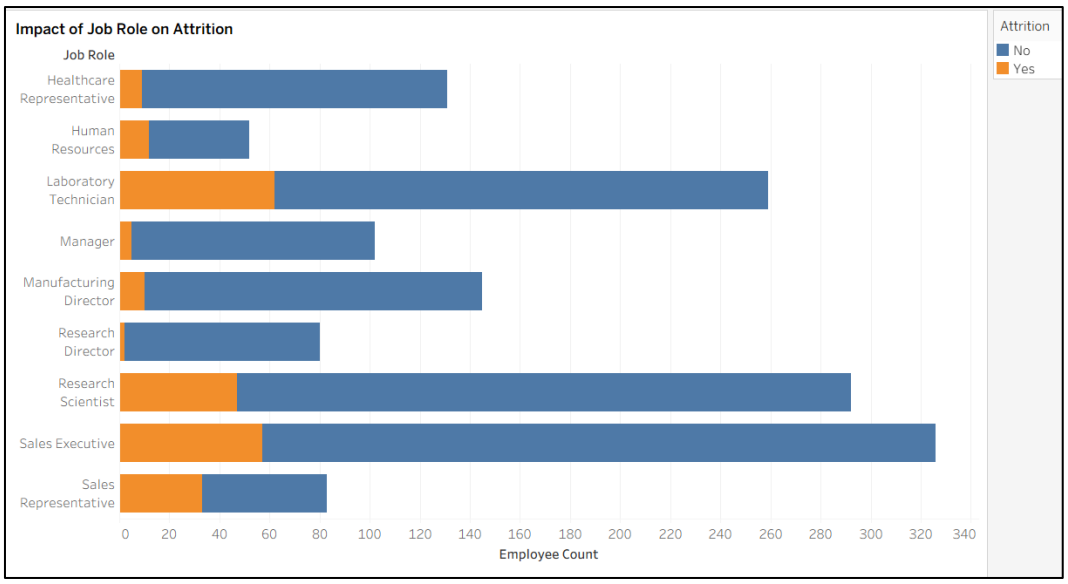

Figure 13: Job Role vs Attrition Chart

The chart reveals that roles such as Laboratory Technician show a disproportionately high attrition rate, with approximately 62 attritions out of 259 employees, corresponding to nearly **23.9%**, the highest among all roles. Sales Executives record around 57 attritions out of 326 employees (**~17.5%**), while Research Scientists show approximately 47 attritions out of 292 employees (**~16.1%**). The visual structure, where certain roles display a noticeably larger attrition segment relative to their total size, indicates that attrition is strongly tied to role-specific conditions such as task repetitiveness, career progression limitations, and performance pressure.

5.2. Operational and Behavioral Drivers

A more direct and statistically dominant factor emerges when examining overtime, where the contrast between categories is both visually and numerically significant.

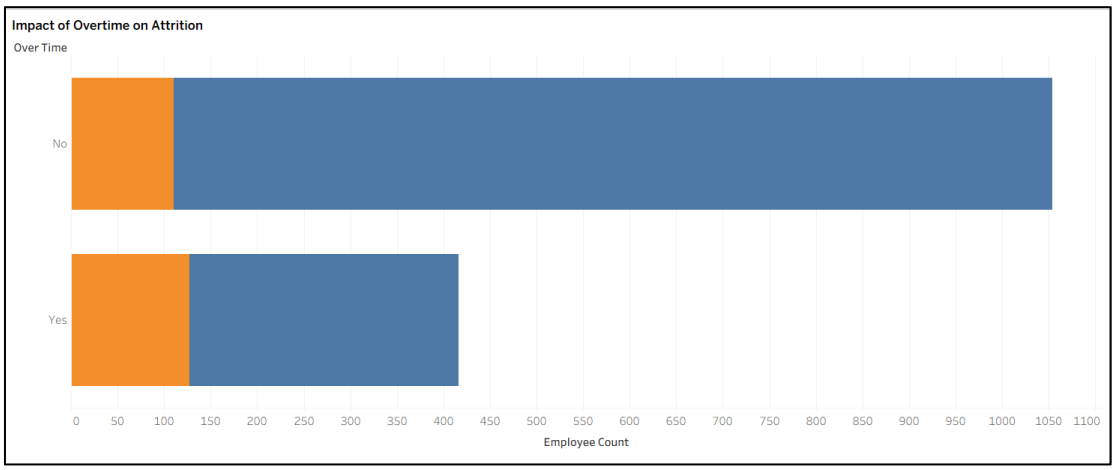

Figure 14: Overtime vs Attrition Chart

The chart compares employees working overtime with those who do not using proportional bar lengths. Among employees working overtime (approximately 416 individuals), around 127 have left, resulting in an attrition rate of about **30.5%**. In comparison, employees not working overtime (approximately 1,054 individuals) show around 110 attritions, corresponding to only **10.4%**. The visual difference is

pronounced, with the attrition segment in the overtime group occupying nearly one-third of the total bar. This indicates that overtime increases the likelihood of attrition by nearly three times, establishing workload intensity as one of the most critical drivers of employee exit.

The influence of work-life balance introduces a more nuanced behavioral dimension, where attrition does not follow a strictly linear trend.

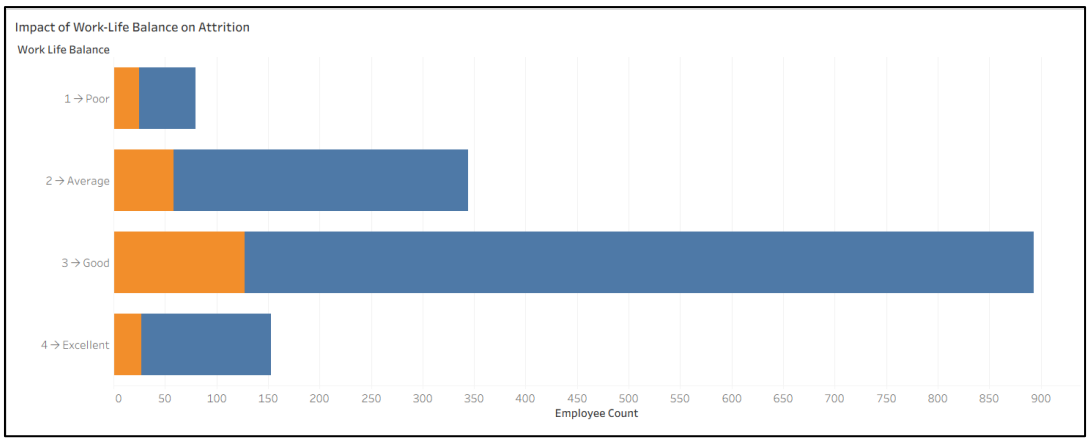

Figure 15: Work-Life Balance vs Attrition Chart

Employees reporting "Average" work-life balance (level 2) exhibit the highest attrition rate at approximately **31.4%**, compared to around **25%** for "Poor" (level 1), **15.5%** for "Good" (level 3), and **20%** for "Excellent" (level 4). Visually, the attrition segment peaks at level 2 before declining, suggesting that prolonged moderate dissatisfaction may be more destabilizing than clearly poor conditions. This indicates that employees are more likely to leave when expectations remain unmet over time rather than during short-term dissatisfaction.

5.3. Economic and Tenure-Based Factors

Income distribution adds a financial dimension to the analysis, highlighting how compensation levels influence attrition behavior.

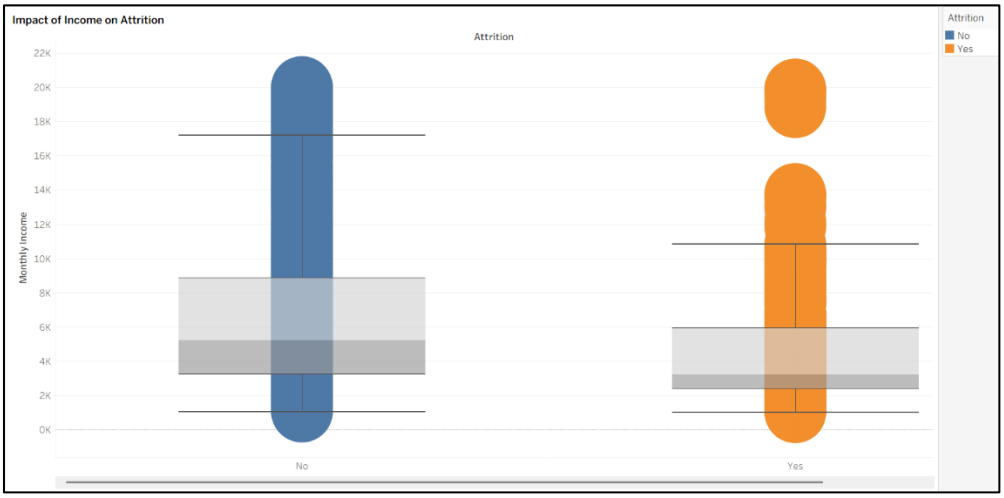

Figure 16: Income Distribution (Box Plot)

The box plot illustrates that employees who left the organization have a median monthly income in the range of approximately **₹3,000–₹4,000**, whereas those who remained have a higher median of around **₹5,000–₹6,000**. The distribution for attrition cases is visibly shifted toward lower income levels, indicating a concentration of exits among lower-paid employees. However, the overlap between the two distributions suggests that income alone does not fully determine attrition, reinforcing the role of combined factors such as workload and job conditions.

Tenure-based analysis further clarifies how attrition evolves over time within the organization.

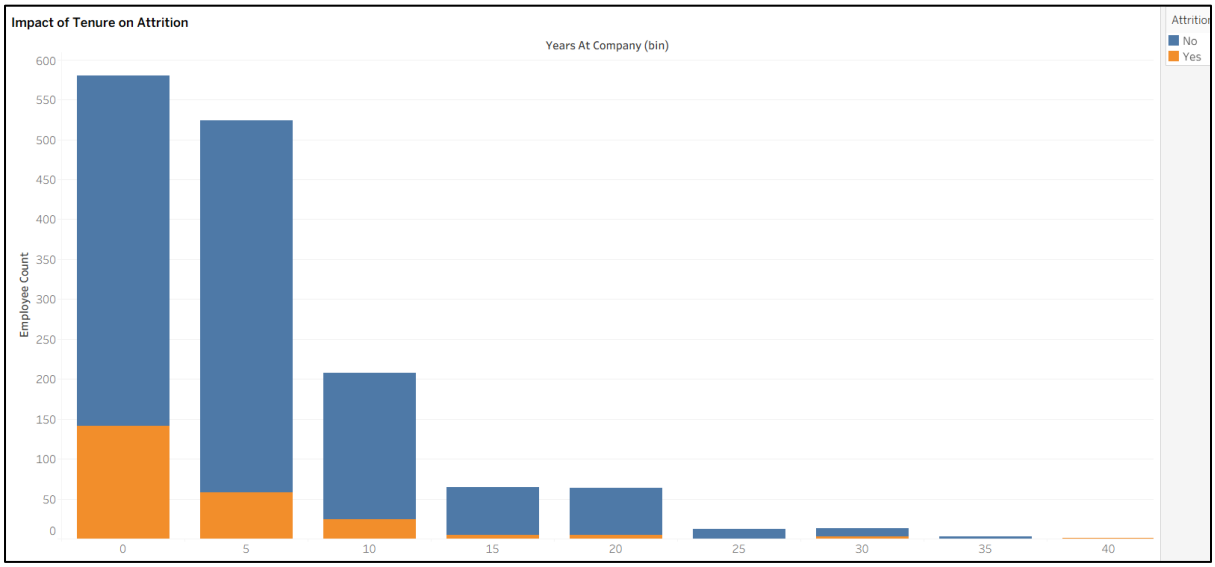

Figure 17: Tenure vs Attrition Chart

By grouping employees into tenure bands, it is observed that the majority of attrition occurs within the first five years. Out of approximately 1,100 employees in the 0–5 year range, around 200 have left, corresponding to an attrition rate of roughly **18%**, which exceeds the overall average. In contrast, employees with 5–10 years of tenure show significantly lower attrition at approximately **8.5%**, while those beyond 10 years exhibit minimal turnover. The visual pattern reflects a steep decline in attrition as tenure increases, indicating that employee retention stabilizes over time.

Across these dimensions, a consistent pattern emerges where attrition is concentrated among employees experiencing higher workload, lower compensation, and early-stage employment, particularly within roles and departments characterized by operational pressure. The integration of these factors demonstrates that attrition is not driven by isolated variables but by the interaction of multiple conditions that collectively increase the probability of employee exit.

6. Filter-Based Dynamic Analysis

6.1. Gender-Based Behavioural Variation

Applying a gender filter reveals how the same attrition system behaves under demographic segmentation rather than aggregate conditions.

Figure 18: Dashboard Filtered for Male Employees

Within the male subset, the workforce reduces to approximately **882 employees**, with around **150 attrition cases**, resulting in an attrition rate close to **17%**, slightly above the overall average of 16.12%. The structural patterns observed earlier remain consistent, but their intensity increases. The overtime chart, in particular, shows a stronger skew, where the proportion of attrition within overtime employees remains close to the previously observed ~30%, but now contributes more significantly to the overall male attrition pool. Department-wise, Sales continues to exhibit elevated attrition, and tenure distribution still shows concentration within the first five years, indicating that early-career instability and workload pressure are amplified within this group.

In contrast, when the dashboard is filtered for female employees, the system presents a slightly moderated version of the same dynamics.

Figure 19: Dashboard Filtered for Female Employees

The female workforce consists of approximately **588 employees**, with around **87 attrition cases**, corresponding to an attrition rate of roughly **15%**, lower than both the male subset and the overall average. While overtime continues to act as a key driver, its relative impact appears less extreme compared to the male subset. Income distribution is also more compact, with fewer extreme low-income cases contributing to attrition. The comparison indicates that while the underlying drivers remain identical across genders, their **magnitude and concentration differ**, suggesting that exposure to high-risk conditions such as overtime and high-pressure roles is unevenly distributed.

6.2. Department-Level Conditional Behaviour

Filtering by department allows the system to be examined in isolation, revealing how different organizational units respond to the same variables.

Figure 20: Dashboard Filtered for Sales Department

Within the Sales department, the attrition rate rises to approximately **20.6%**, and the effect of overtime becomes more concentrated. A larger proportion of employees fall into overtime categories, and the attrition segment within this group expands accordingly. Work-life balance levels also tend to cluster around "Average" and "Poor," reinforcing the combined effect of workload and moderate dissatisfaction. This indicates that Sales operates as a **high-pressure subsystem**, where multiple risk factors—overtime, performance expectations, and lower perceived balance—align simultaneously.

When the same dashboard is filtered for Research & Development, the behavior shifts noticeably.

Figure 21:Dashboard Filtered for R&D Department

Despite having the largest workforce (~961 employees), the attrition rate is lower at approximately **13.8%**, and the impact of overtime is less uniformly distributed. Attrition appears more localized within specific roles rather than spread across the department. Income variation is broader, but does not produce a strong direct correlation with attrition in this segment. This indicates that R&D is structurally stable at the macro level, with **localized inefficiencies rather than systemic stress**.

6.3. Role-Specific Conditional Patterns

Further refinement through job role filters exposes how attrition behaves at the most granular level of organizational structure.

Figure 22: Dashboard Filtered for Laboratory Technician Role

For Laboratory Technicians, the attrition rate reaches approximately **23.9%**, significantly above the overall average. The income distribution for this role is concentrated in lower ranges, and while overtime contributes, it does not dominate as strongly as in Sales. Instead, the chart suggests that attrition is driven by a combination of **limited growth opportunities, repetitive work, and moderate compensation**, making this a structurally vulnerable role.

In contrast, when observing managerial roles:

Figure 23: Dashboard Filtered for Manager Role

Attrition levels drop significantly, often remaining below **10%**, with higher income ranges and minimal overtime influence. The visual distribution shows a stable retention pattern, where both workload and compensation align favorably. This demonstrates how increased autonomy and financial incentives act as stabilizing factors within the system.

Across these filtered views, the underlying system remains consistent, but its expression changes depending on the segment being observed. The same variables—overtime, income, tenure, and role—continue to drive attrition, but their **relative weight shifts across gender, department, and job role**. This confirms that attrition is not governed by isolated conditions but by **context-dependent interactions**, where risk intensifies when multiple factors converge within specific segments of the workforce.

7. Workforce Risk Segmentation Model

7.1. Model Construction and Rationale

To move beyond descriptive analysis, the workforce is segmented into risk categories based on likelihood of attrition. The overall attrition rate (~16.12%) indicates the scale of the problem but does not identify *where* intervention is most needed. Risk segmentation addresses this by classifying employees based on their exposure to key attrition drivers, enabling targeted analysis rather than uniform assumptions.

The model is constructed using variables that showed strong influence in prior analysis—**overtime status, work-life balance, monthly income, and years at company**. These variables collectively represent workload, well-being, compensation, and organizational attachment, and are used in combination to approximate attrition risk rather than in isolation.

7.2. Classification Logic

The segmentation is implemented using a calculated field in Tableau:

```
IF [OverTime] = "Yes"
   AND [WorkLifeBalance] <= 2
   AND [MonthlyIncome] < 4000
THEN "High Risk"

ELSEIF [OverTime] = "Yes"
   OR [WorkLifeBalance] = 2
THEN "Medium Risk"

ELSE "Low Risk"
END
```

Figure 24: Risk Segmentation Formula

Employees exposed to multiple adverse conditions are classified as **High Risk**, those with partial exposure as **Medium Risk**, and the remaining as **Low Risk**, reflecting the compounding nature of attrition drivers.

7.3. Distribution and Interpretation

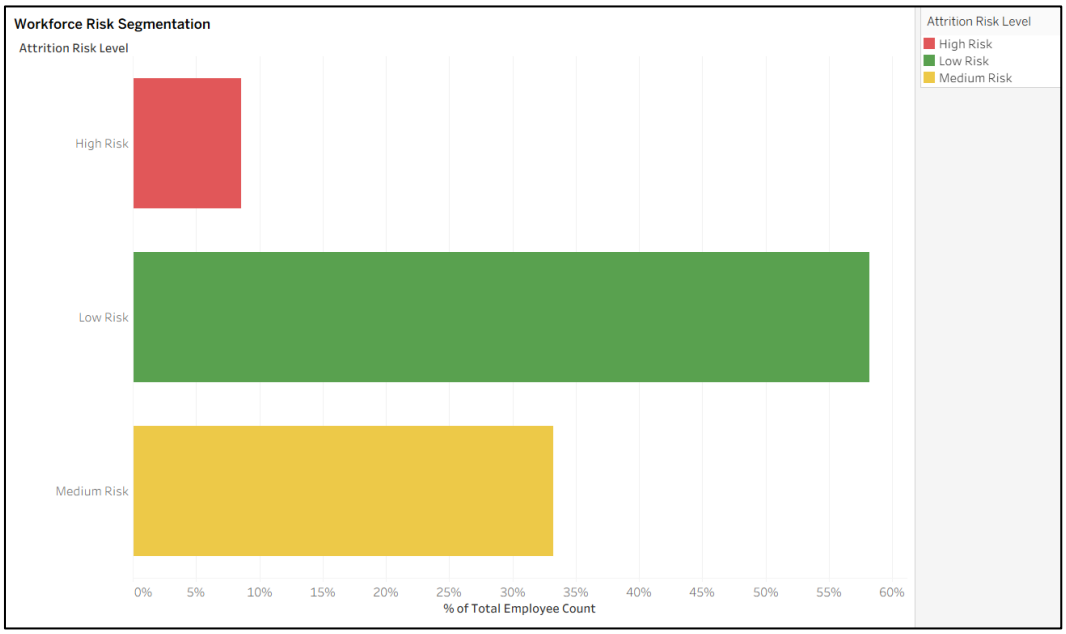

Figure 25: Risk Segmentation Chart

The resulting distribution shows approximately **58% Low Risk (~850 employees)**, **33% Medium Risk (~480 employees)**, and **9% High Risk (~130–140 employees)**. Although the High Risk group is the smallest, it represents the most critical segment due to its elevated probability of attrition.

This concentration implies that targeted intervention within this group—such as reducing overtime or improving compensation—can produce a disproportionately large impact on overall attrition. The Medium Risk group, on the other hand, represents a transitional segment where early corrective measures can prevent escalation.

8. Integrated System Interpretation

8.1. Convergence of Early Tenure, Workload, and Income

When individual variables are examined together, attrition emerges as a concentrated outcome rather than a distributed one. The overall attrition rate of **16.12% (237 out of 1,470 employees)** is largely driven by employees within the **0–5 year tenure group (~1,100 employees)**, where attrition rises to approximately **18% (~200 exits)**. This baseline risk increases sharply when combined with overtime exposure. Employees working overtime exhibit an attrition rate of approximately **30.5%**, compared to only **10.4%** for those who do not, indicating nearly a threefold increase in exit probability.

When income is layered onto this interaction, the system becomes more defined. Employees with median monthly income in the range of **₹3,000–₹4,000** are disproportionately represented among attrition cases, while retained employees cluster closer to **₹5,000–₹6,000**. Within the early-tenure group, lower income reduces tolerance to sustained workload, creating a compounded condition where **early tenure + overtime + low income** forms the most unstable segment. This group aligns closely with the **High-Risk category (~9% of workforce, ~130–140 employees)**, which contributes disproportionately to total attrition.

8.2. Role Pressure and Work-Life Balance Interaction

A parallel system operates through the interaction of job role and work-life balance. Roles such as Laboratory Technician (**~23.9% attrition**) and Sales Executive (**~17.5% attrition**) demonstrate that attrition is influenced not only by workload but by the structural nature of the role itself. Within these roles, work-life balance ratings frequently concentrate around "Average" (level 2), which shows the highest attrition rate of approximately **31.4%**, exceeding both "Poor" (~25%) and "Good" (~15.5%) levels.

This non-linear pattern indicates that attrition is driven less by extreme dissatisfaction and more by sustained moderate strain. Employees in high-pressure roles with "Average" work-life balance are not in crisis, but they are consistently strained without improvement, making exit more likely over time. In contrast, roles with higher autonomy and compensation, such as managerial positions, show attrition rates below **10%**, reinforcing the stabilizing effect of control and financial incentive.

8.3. System-Level Behavior and Predictive Insight

When these interactions are combined, attrition can be understood as a system where multiple conditions align to produce high exit probability. The highest-risk profile consists of employees who are early in tenure, working overtime, earning lower income, and operating in high-pressure roles with moderate work-life balance. This group, while representing less than **10% of the workforce**, accounts for a disproportionate share of attrition outcomes.

From a predictive perspective, the system suggests that future attrition will continue to originate from this cluster unless structural conditions change. Reducing overtime exposure within early-tenure

employees alone could potentially lower overall attrition from **~16% to approximately 12–13%**, based on observed differences in attrition rates. Similarly, improving compensation within the lowest income band or shifting work-life balance from "Average" to "Good" could reduce attrition probability within affected segments by **10–15 percentage points**.

This interconnected behavior confirms that attrition is not driven by isolated variables but by **compounding interactions**, where risk intensifies as multiple conditions overlap. Understanding the system at this level allows for targeted intervention strategies that address root causes rather than surface-level symptoms, making workforce management both more efficient and more effective.

9. Conclusion

The analysis confirms that employee attrition is a structured and measurable outcome, not random variation. With **237 exits out of 1,470 employees**, the overall attrition rate stands at **16.12%**, but these average masks strong concentration within specific workforce segments. Employees in the **0–5 year tenure group account for nearly 200 exits**, with attrition around **18%**, which increases sharply under workload pressure. Among employees working overtime, attrition rises to approximately **30.5%**, compared to **10.4%** for those not working overtime, indicating nearly a threefold increase in exit probability.

Income further differentiates outcomes, with employees who leave concentrated around a median monthly income of **₹3,000–₹4,000**, while retained employees cluster closer to **₹5,000–₹6,000**. Role-level variation reinforces this pattern, with positions such as Laboratory Technician reaching attrition levels of approximately **23.9%**, compared to less than **10%** in more stable managerial roles. Work-life balance also shows a measurable effect, with "Average" conditions producing the highest attrition at approximately **31.4%**, compared to **15.5%** under "Good" conditions.

When combined, these factors define a high-risk segment representing approximately **9% of the workforce, or 130–140 employees**, where attrition probability is significantly elevated. This group contributes disproportionately to total attrition, indicating that turnover is driven by concentrated conditions rather than uniform distribution.

The Tableau dashboard enables these relationships to be identified, compared, and filtered across dimensions such as gender, department, and role, allowing the same dataset to be analyzed under multiple conditions without restructuring the model. This supports precise, data-driven decision making, where interventions can be directed toward high-impact segments. Based on observed differences, reducing overtime exposure alone has the potential to lower overall attrition from approximately **16% to near 12–13%**, while targeted improvements in compensation and work conditions within early-tenure employees could further reduce attrition by **10–15 percentage points** within affected groups.

The findings demonstrate that attrition can be quantified, segmented, and systematically reduced when analyzed as an interconnected system rather than isolated variables.